BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

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Microprocessors and Embedded Systems Laboratory

Final Project Report

Section: C1 Group: 02

IoT-Based Aquaculture Monitoring System

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Academic Honesty Statement:

IMPORTANT! Please carefully read and sign the Academic Honesty Statement, below. Type the student ID and name, and put your signature. *You will not receive credit for this project experiment unless this statement is signed in the presence of your lab instructor.*

"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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1 Abstract

The production of aquatic organisms, or aquaculture, is essential to supplying the world's growing seafood demand. Nonetheless, there are a lot of obstacles in the way of keeping aquaculture operations sustainable and providing ideal circumstances for aquatic life. In this regard, the incorporation of Internet of Things (IoT) technology is a viable way to control and monitor aquaculture ecosystems in real time.

The goal of this project is to improve aquaculture operations' sustainability, productivity, and efficiency by implementing an Internet of Things-based aquaculture monitoring system. Key parameters like water level, temperature, pH levels, and feeding patterns are monitored by the system through a network of sensors, actuators, and data analytics tools. These sensors are placed throughout tanks, cages, and other aquaculture facilities to enable constant environmental condition monitoring.

2 Introduction

The management of aquatic environments for sustainable production is a complicated engineering challenge that is addressed by the development of an IoT-based aquaculture monitoring system. Aquaculture operations demand careful control over various dynamic factors such as water level, temperature, and pH levels, posing intricate challenges for traditional management methods. These difficulties are made even more difficult by the requirement for effective resource management and the possibility of disease outbreaks, which can have catastrophic consequences for aquaculture results. Alternative options like manual monitoring are constrained by their labor-intensive nature and incapacity to offer real-time insights. While automated monitoring systems can be an improvement, they might not be as sophisticated as what is needed for full and proactive management. Selecting proper sensors, setting the parameters and interfacing the sensor data with the IoT cloud made this project more complex.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

Our project contributes to the automation of aquaculture as well as make things easier for the fish farmers of our country. As in Bangladesh we have a lots of water bodies and fish firming is one of the most important job to fulfill the need of protein. Many of the fish firms are dependent on the manual monitoring and traditional decision making systems. The project aims to enhance efficiency, optimize resource utilization, and ultimately improve the overall productivity and sustainability of aquaculture operations through data-driven insights and proactive management.

3.1.2 Literature Review

We have searched for a good number of papers related to this project and studied them for better idea. The papers are attached in the reference section.

3.1.3 Formulation of Problem

In our project we have planned our work in two major parts:

Obtaining sensor data: At first, we executed the part of getting real-time data from the sensors. The temperature, water level and the pH level values were obtained from the sensors. We had to calibrate those sensor values to get an understandable value to make further decisions and monitor the system.

Taking decision to make actions: After we got the parameter values from the sensor we processed the data and set threshold to make decisions. The decision also include an action according to the value. We also executed the plan to make an automated feeding system.

3.1.4 Analysis

We have analyzed the problem and found out possible challenges in this work and searched for the resources to overcome those problems.

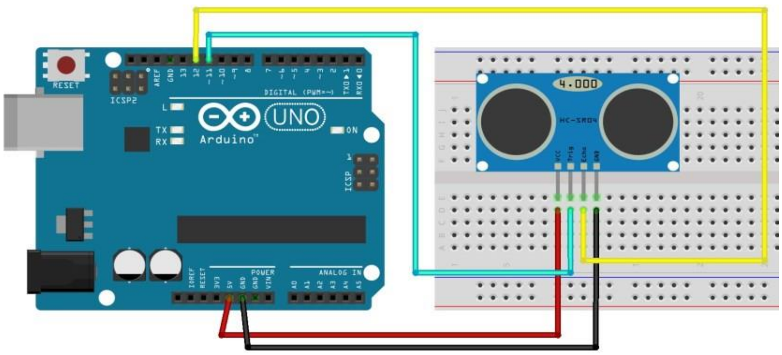
3.2 Design Method (PO(a))

We have used 3 sensors and one servo motor.

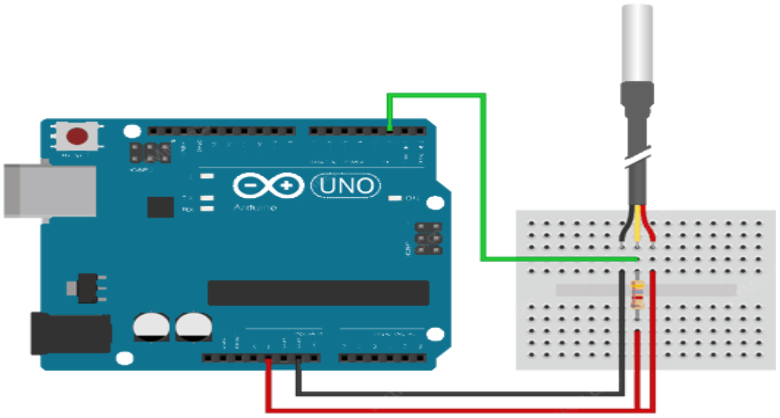
1. Waterproof temperature sensor(DS18B20): to collect the temperature data of the pond.
2. Water level sensor: to collect the data of water level.
3. pH sensor: to collect the pH level data.
4. Servo motor: to implement the feeding system.
5. Arduino UNO
6. ESP32

3.3Circuit Diagram

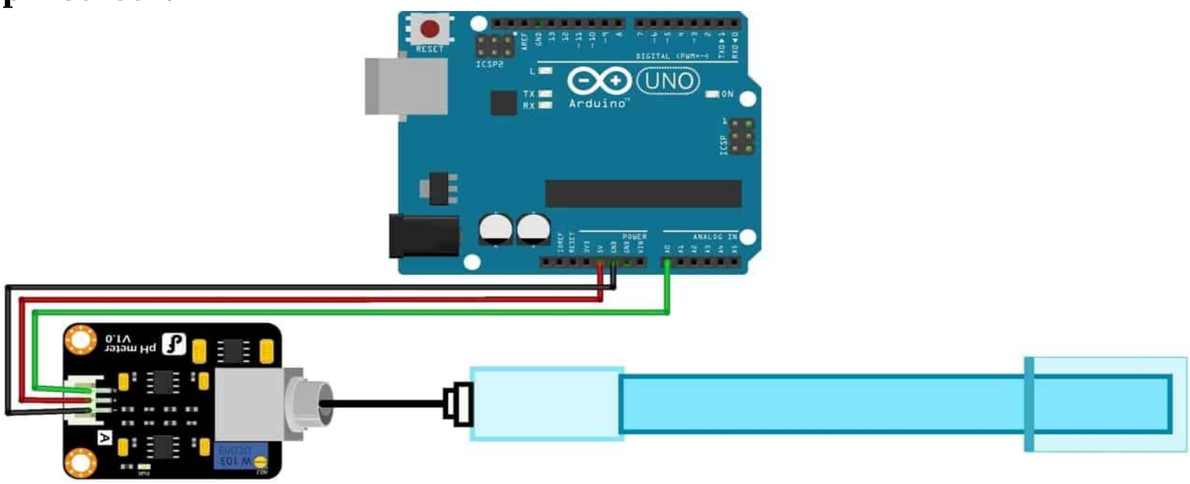
1. **Water level sensor:**



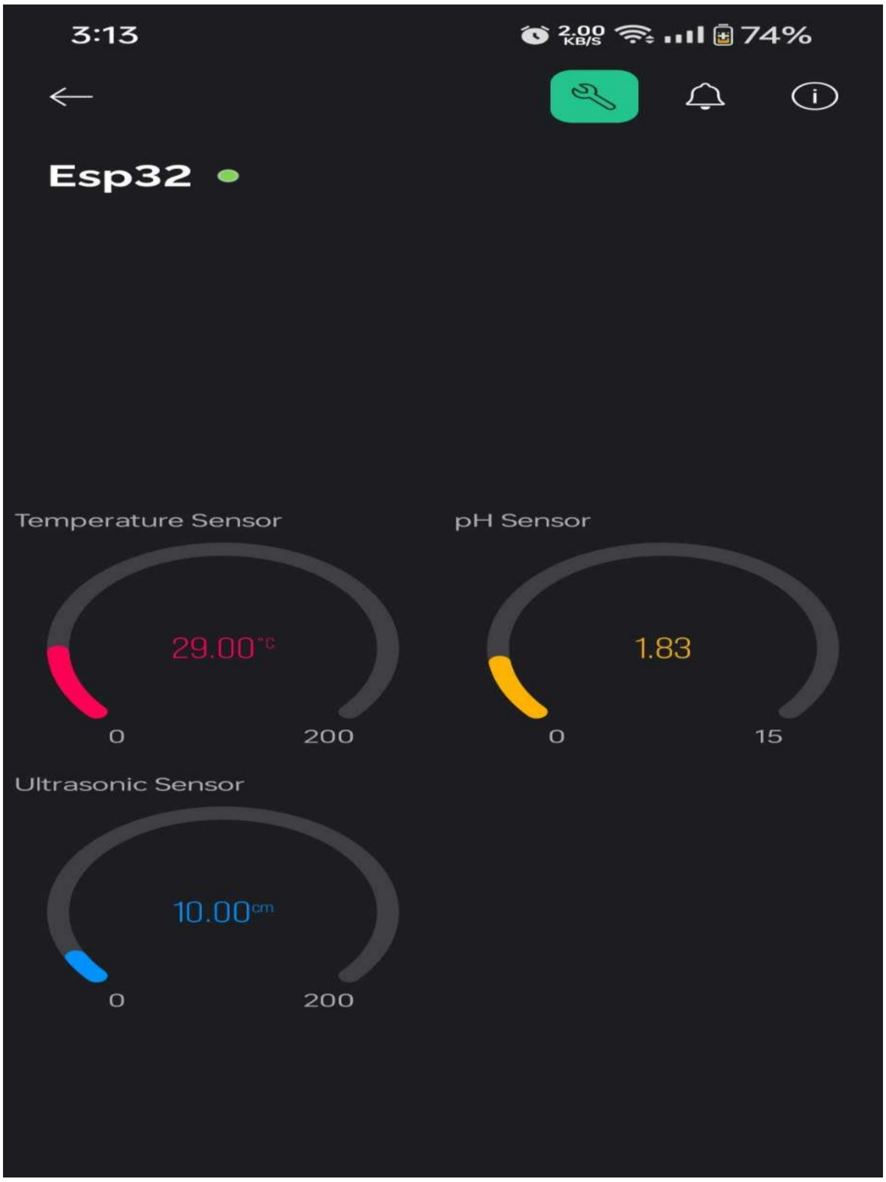
Waterproof temperature sensor:



pH sensor:



We have integrated this circuits, interfaced them with ESP32 and monitored the sensor data in the Blynk app.

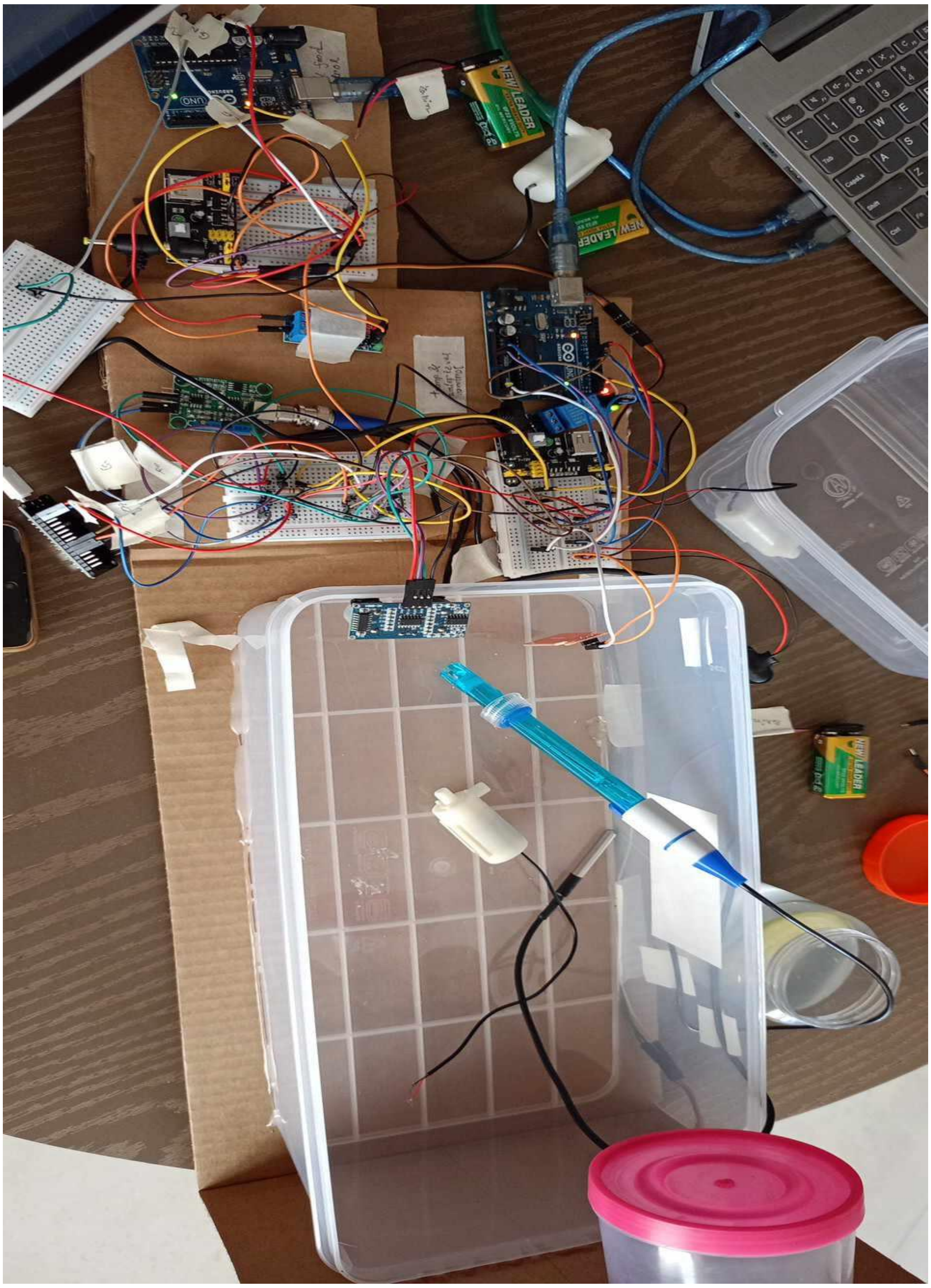


4 Implementation

4.1 Description

This is our full setup pic.

1. If temperature is greater than 32 degree celcius then the pump will supply water.
2. If pH is less than 5 then another pump will supply lime water.
3. If water level is less than half of the pond water then the third pump will supply water.
4. The servo motor will rotate the feeding pot and add feed to the pond.



5 Design Analysis and Evaluation

5.1 Novelty

There are total four systems in our project which are temperature balancing module, pH balancing module, water level balancing module and feeding system. The novelty of our project is that our project not only monitor the parameters like pH level or temperature but also take actions to balance these parameters when they go beyond the threshold we set. All the hardware setup and integration of the modules as well as the data processing were done by the team members.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

This project can be useful for public health and safety in different ways:

IoT based aquaculture monitoring system can also be used to monitor water quality as well as maintain health safety for human.
This project also ensures monitoring and feeding at a fishpond from a distant place. So the farmers don’t need to visit the pond regularly. This facility also make possible to establish firms at remote area.
This project can be further used for unpredictable situations such as flood or cyclone.
The equipment used in the project are free from health hazards and very safe to use.
Our project also ensures food safety with diseases free fish by maintaining the environment of fish firm.

5.2.2 Considerations to environment

IoT based aquaculture monitoring system is a very environment friendly project under the following considerations:
We have designed our project with energy efficient sensors and equipment to ensure low energy consumption, reduce water consumption and waste generation.
This project prevent fish diseases by maintaining water quality. As a result use of medicine and chemicals to the fish firm is reduced and the ecosystem of the pondis protected from the harmful effects of chemicals.
This project is designed to ensure limited amount of feed for the fish by the feeding system which helps to prevent from water quality degradation by overfeeding.
The systems of our project can also be used to monitor the water quality of the surrounding water bodies to examine the impact of a fish firm.

5.2.3 Considerations to cultural and societal needs

Our project does not violate the social and cultural boundaries with the considerations in mind:

The local community, where the fish farm is situated, respects and acknowledges its cultural customs and traditions. Through this project, it will be ensured that the farm's operations and the monitoring system adhere to cultural norms and values. With the facilities of this project the local community will be encouraged for fish firming and get involved into this. Also it offers employment to the people for the management of this project.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

The design of our project was quite difficult as we have used three sensors, three pumps and a servo motor. The pH sensor gave us a voltage value which was calculated to a readable value by several calibration. At first we tried to give power to the pumps from battery and then we used a module to supply the pumps from ac power supply. But this was not useful as it could not supply enough power to the pumps. So we again switched to the battery. We used Arduino Uno and ESP32 to collect sensor data and uploaded the code in the IoT cloud to monitor the real time data. we used a servo motor for the feeding system to rotate the feeding jar after a certain time. The control part was very difficult to implement and we had to make a lot of trial and error to overcome that issue.

5.3.2 Data Collection

We have collected data from the sensor and processed for further work.

5.3.3 Results and Analysis

After collecting the sensor data the system compares that data with the threshold and take actions according to it.
If the water level is low, the pump associated in this module supplies water from the reservoir to the pond.
If the temperature of the water is high, the pump supplies water to reduce temperature of the pond.
If the pH level is lower than 5 which means the water is acidic, a pump supplies lime water to balance pH level. When the pH level is higher then 8.5, the pump stops supplying lime water.

5.3.4 Interpretation and Conclusions on Data

We were able to monitor the data in the IoT cloud and the actions were taken accordingly. However the success of this project fully depends on proper hardware connection and a stable wi-fi connection.

5.4 Limitations of Tools (PO(e))

The limitations we faced during the implementation of our project are as follows:

1. We cannot measure the amount of lime water added to the pond because the flowrate of pump cannot be exactly known. We have run the pump for a few second and then stop and measure the value and then took the next decision.

2. We could not use the ac to dc module as it could not supply enough power to the pumps. As a result, we had to implement the project using 9V battery.

3. The pH sensor sometimes gives garbage value which is a vital limitation of our project but we could not overcome this by affording more efficient pH sensor because of our budget limitations.

4. The water level sensor sometime gives wrong data if any water droplet adheres to its body at a higher level than the actual water level. We tried to overcome this by using ultrasonic sensor.

5. We have to calibrate the pH sensor data every time we start to run the project.

5.5Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

1. Automation of fish firm using this project may increase the production and efficiency of fish firm which contributes to the economy of a society or a community. Also the increment in production will meet the need of nutrition and reduce the pressure on imported seafood.
2. people may get involved into this project for further development and

maintenance of this work which may result into employment opportunities. Also, people may develop their skills for fish firming through this project.

3. Many communities are stick to their traditional way of fish firming. This project may affect their traditional practices. In this case, special consideration should be given to introduce them to this IoT project as a complement to their traditional way fish firming respecting their cultural norms
4. This project can raise awareness among the people to accept improved technologies to increase efficiency.

5.5.2 Assessment of Health and Safety Issues

1. As the water quality is continuously monitored and immediate actions are taken according to any abnormalities, this project prevents any unexpected condition to occur ensuring safety.
2. Maintaining water quality this project prevents fish diseases and spread of the diseases among the consumers. Thus, health and food safety are maintained.
3. Fishes grown in clean and well-maintained water are obviously healthy for consumer and contains less harmful elements than the fish grown in dirty water.
4. Maintaining water quality helps to reduce the use of harmful chemical and medications in the fish firm thus helps in improving health and safety issues.

5.5.3 Assessment of Legal Issues

This project does not support any illegal things. So there is no observable impact on legal issues.

5.6 Sustainability Evaluation (PO(g))

The "IoT-based Aquaculture Monitoring System" uses real-time data collecting and analysis to improve fish farming's sustainability and efficiency. This is a brief assessment of its sustainability in terms of social, economic, and environmental factors:

Environmental Benefits:

1. Minimizes waste and environmental impact while optimizing the usage of resources.
2. Monitors water quality to protect aquatic ecosystems.
3. Increases energy efficiency, which may help to lower carbon emissions.

Environmental Drawbacks:

1. Generates electronic waste from IoT devices.
2. Consumes more energy even though it is intended to be energy-efficient.

Economic Benefits:

1. reduces operating expenses through feed, energy, and water treatment optimization.
2. Increases profitability through better yields and growth rates.
3. Improves risk management by promptly recognizing and resolving problems.

Economic Drawbacks:

1. Requires significant initial investment for setup.
2. Incurs continuous maintenance and employee training expenditures.

Social Benefits:

1. Creates tech jobs in rural and coastal areas.
2. Encourages the use of sustainable aquaculture methods in communities.

Social Drawbacks:

1. Gives a skills challenge for local communities unfamiliar with IoT technologies.
2. Risks of growing cost differences between large and small-scale businesses.

5.7 Ethical Issues (PO(h))

Environmental management: We tried to minimize the environmental harm by using materials which can be recycled. We have also reused some component like Arduino and jumper wires. The IoT devices were chosen based on their energy efficiency and longevity to minimize electronic waste.

Economic Accessibility: Developing scalable solutions and training programs will make the technology accessible to operators of all sizes, ensuring no one is left behind.

Social responsibility: We tried to make this project a good scope for community people and things are easier to get involved. A challenge in this regard is the technological lacking of people. We can overcome this by arranging training sessions in collaboration with the government.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

1. **Rifah Nanzeeba**

   1.Finding project idea and resources.

   2.Writing code for pH sensor and water level sensor and debugging.

   3. Preparing project development slide and project report.

2. **Rabeya Salam Munira**

   1. Finding project idea and resources.
   2. Finding resources for pH sensor and water level sensor and writing code.
   3. Helping in hardware implementation and circuit construction.

3. **Monideepa Kundu (Sinthi)**

   1. Interfacing ESP32 with the IoT cloud.
   2. Finding resources for Temperature sensor and feeding system and writing code.
   3. Preparing project development slide

4. **Most. Noor Afroz Rimu**

   1. Helping in interfacing ESP32 with IoT cloud.
   2. Helping in hardware implementation and temperature sensor connection.
   3. Writing project report and preparing project development slide.

6.2 Mode of TeamWork

We worked as a team for this project. We had divided the works among all team members and proceeded accordingly. We had several offline meetings to discuss the problems we faced and tried to solve them. All the members of our team were respectful to each other.

6.3 Diversity Statement of Team

We really had a diversity of team members. Two of us were hall residents and two of us were attached students. So we had to schedule our offline meetings considering everyone's situations. Also all the members were not from the same major group. So we had to work on weekends. As we have mentioned in the previous section that we had divided our work so it was not much difficult for us to have a diversified journey during this project.

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

Here we present our IoT based aquaculture monitoring system. This system uses sensors to collect data from a fishpond and after processing the data takes necessary actions to avoid any abnormality in the fishpond. This project can help you to improve your fish firming idea and increase efficiency and productivity. We have used three different sensors: temperature sensor, pH sensor, water level sensor. This is a very cost effective and low energy project which will enhance your fish firm's quality helping with the monitoring system.

8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

Equipment	Quantity	Price(tk)
pH sensor + module	1	1900
5V pump	3	180
ESP32	1	370
5V relay	3	150
Water level sensor	1	60
Servo motor	1	250
Temperature sensor	1	120
Power supply module	2	400
Breadboard	5	300
9V Battery	4	280
Pipe, jumper wires, others		200
Total		4210

8.2 Calculation of Per Unit Cost of Prototype

We have to account for the transportation cost of materials and also the cost of buying things that were not used in the final product. The per unit cost of prototype amounts to about 4000 tk.

8.3 Calculation of Per Unit Cost of Mass-Produced Unit

Total cost of our finished product is about 3500 tk. Cost of mass production can be a little less than this.

8.4 Timeline of Project Implementation

Add a Gantt chart here

9 Future Work (PO(l))

In the future, we hope to create solutions for some of the issues that we were unable to address in the end. Some of these have been outlined below:
1. Other sensors like salinity sensor, oxygen level sensor etc can be added to have a much better water quality control.

2. In case of action making decision the threshold and the decision making parameters can be more accurate by using high quality sensors.

10 References

https://www.researchtrend.net/ijet/pdf/8%20IoT%20Based%20Smart%20Fish%20Farming%20Aquaculture%20Monitoring%20System%20Sohail%20Karim%203461.pdf

https://www.researchgate.net/publication/366098813_Internet_of_Things_IoT_Based_Aquaculture_Monitoring_System